Lindsey Ma

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Education

Columbia University, M.S. in Statistics, GPA 3.7 *Sep 2024 – May 2026*

Coursework: Deep Learning, Advanced Machine Learning, Statistical Analysis of Neural Data, Robot Learning

Renmin University of China, B.A. in Economics, GPA 3.8 *Sep 2019 – Jun 2023*

Experiences

Software Engineer Intern - Applied ML for Plant Identification *May 2026 – Present*
Arborphy AI LLC *New York*

- Designed a **dual-encoder architecture** combining EfficientNet and BioCLIP via cross-attention feature fusion to integrate organism- and species-level representations while mitigating negative transfer.
- Diagnosed **representation** failures using frozen-embedding clustering, linear probes, and rotation-based local projections; active learning-guided fine-tuning improved top-1 accuracy by 6% over zero-shot CLIP.
- Distilled a lightweight student model for offline iOS inference, with cloud fallback for ambiguous cases.

Software Developer - Real-Time Vision for Robotics *May 2025 – Dec 2025*
InnoX Academy Incubator Robotics Lab *Shenzhen*

- Built a 30 FPS real-time vision pipeline on Raspberry Pi for a badminton training robot, integrating pose estimation, shuttlecock tracking, and court positioning to drive adaptive serving and technique feedback.
- Deployed a local LLM voice interface with Ollama and onboard audio for hands-free robot control. See code.

Research Experience

Research Assistant - Self-Supervised Learning for Neural Imaging *Jan 2026 – May 2026*
Center for Theoretical Neuroscience, Columbia Zuckerman Institute *New York*

- Designed and trained a **self-supervised blind-spot neural network** in PyTorch for denoising two-photon calcium-imaging movies under temporally correlated noise.
- Conducted systematic ablations of masking strategies, receptive fields, and preprocessing across five real and simulated imaging datasets, evaluating latent-signal recovery and neuronal morphology preservation.
- Rebuilt the lab's PMD preprocessing pipeline, reducing residual video energy by 20% and downstream decoding variance by 7% versus baseline. See paper and code.

Selected Projects & Activities

Hackathon Team Lead - 2nd Place, XRPL ($5K); 3rd Place, Stellar ($3K) *Sep 2025 – Oct 2025*

- Led a 4-person team to ship two fully deployed projects in separate 24-hour blockchain hackathons; owned backend architecture, live on-chain transaction pipelines, integrations, and end-to-end deployment. See code.

FlyWire Connectome Analysis - Computational Statistics Project *May 2025*

- Applied **Bayesian stochastic block models** to FlyWire whole-brain connectome to infer cell-type organization; evaluated community structure with posterior predictive checks and embedding analysis. See paper.

Columbia AI Alignment Club - Technical Fellow *Feb 2025 – May 2025*

- Reproduced Anthropic's **hallucination-circuit analysis** on Qwen-8B using transcoder-based attribution graphs; applied feature-guided activation steering to induce reproducible refusal-to-hallucination shifts.

Brain Computer Interface Lab - EEG Decoding Project *January 2025*

- Trained a linear discriminant analysis (LDA) classifier on real-time EEG spatial features extracted via Common Spatial Pattern to decode left- versus right-hand motor imagery, achieving 80% mean accuracy.

Technical Skills

Fluent in: Python, SQL, R **ML:** PyTorch, Hugging Face, scikit-learn
Tools: Linux, Git, Docker, FastAPI, vLLM, Neo4j